LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Quick Revision: Force and Laws of Motion

Key terms

Term	Meaning
Balanced forces	Equal and opposite forces; no change in the state of motion
Unbalanced force	A net force; it changes speed or direction (causes acceleration)
Inertia	The tendency of a body to resist a change in its state of rest or motion. Mass is a measure of inertia.
Momentum (p)	p = mv. A vector, in kg m/s
1 newton	The force that gives a 1 kg mass an acceleration of 1 m/s^2

Newton's laws

- **First law:** a body stays at rest, or in uniform motion in a straight line, unless an unbalanced force acts on it (law of inertia).
- **Second law:** the rate of change of momentum is proportional to the applied force, in the direction of the force: **F = ma**.
- **Third law:** to every action there is an equal and opposite reaction. **Action and reaction act on different bodies**, so they never cancel.

$F = ma = (mv - mu)/t \cdot p = mv$
Conservation of momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ (total momentum before = after, if no external force)

Everyday examples

- Passengers lurch forward when a bus stops suddenly; beating a carpet removes dust: **inertia**.
- A fielder pulls the hands back while catching: more time, so a smaller force (second law).
- Gun recoil, rockets, swimming, walking: **third law**. Seat belts reduce injury by slowing us down over a longer time.

Common mistakes

- Using grams instead of kg, or km/h instead of m/s, in F = ma.
- Saying action and reaction cancel. They act on different objects.

Quick check

1. A 1000 kg car moving at 20 m/s is stopped in 5 s. Find the force applied by the brakes.

2. Why does a gun recoil when a bullet is fired?